

Peter Van Katwyk

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Professional Summary

PhD candidate in Scientific Machine Learning with 5+ years of experience building and deploying scalable ML systems across NASA, Meta, and academia. Strong background in deep learning, cloud-native ML (AWS, PyTorch), uncertainty quantification, and temporal and geospatial modeling. Skilled at translating complex research into production-ready tools that drive real-world impact.

Education

Brown University, PhD in Scientific Machine Learning Sep 2021 – May 2026

- **Advisor:** Karianne Bergen, **Dissertation:** AI for Climate, Bayesian NNs for Projecting Future Sea Level
- National Science Foundation GRFP Fellow

Brigham Young University, BS in Geological Sciences, Minor in Applied Statistics Sep 2017 – May 2021

Technical Skills

Programming: Python, SQL, Bash, MATLAB, R, Julia, C++

ML/AI: PyTorch, TensorFlow, Bayesian modeling, GenAI, Spatial/Temporal Modeling

Data & Tools: AWS (SageMaker, Lambda, API Gateway, DynamoDB, etc.), Docker, PostgreSQL, Git/Github

DevOps/MLOps: WandB, Flask, DVC, MLFlow, Containerization, Cloud Deployment, CI/CD

Experience

Research Scientist (PhD), NSF GRFP Fellow, Brown University – Providence, RI Sep 2021 - Present

- Built Bayesian deep learning emulators for geospatial ice sheet modeling that cut sea level projection error by **80%** and reduced inference time by **10x**, enabling faster, more reliable decision-making tools for climate planning
- Developed and maintained **ise**, a scalable production-grade open-source Python library for end-to-end data preprocessing, training, and evaluation of scientific ML models, integrated with PyTorch and HPC/GPU capabilities
- Collaborated across scientific and ML disciplines, and advanced methods for uncertainty quantification and published four first-author papers (*TMLR*, *GMD*, *JAMES*, *Nature Comm.*) with co-authorships in *Nature* and *The Cryosphere*

Machine Learning Engineer (Contract), Meta – Remote Aug 2020 – May 2025

- Led design, prototyping, and deployment of cloud-scale infrastructure for a geospatial ML platform supporting large-scale fiber network expansion
- Built and productionized **4** time-series and computer vision models and LLM-driven geospatial models for automated soil classification, permitting, and reporting—used by **150+** engineering teams at Meta and Vermeer
- Conducted training sessions on Data Science and ML best practices and deployment, improving adoption and maintainability across Meta and Vermeer teams

Data Scientist Intern, NASA – Mountain View, CA Summer 2024

- Conceived and developed a Reinforcement Learning framework, **aiatc**, in PyTorch for geospatial AI-based air traffic control, including collision avoidance, dynamic weather rerouting, and sector-aware path planning
- Delivered **6%** reduction in flight times during simulated weather disruptions; framework maintained over **99.7%** routing success rate in simulation tests, demonstrating robustness under real-world conditions
- Presented work at NASA-wide conference and earned *Outstanding Intern Presentation* award

Publications

Selected Publications

- **Van Katwyk et al.** (2025). Rewiring climate modeling with machine learning emulators. *Nature Communications Earth & Environment* (under review).
- **Van Katwyk & Bergen** (2025). HybridFlow: Quantification of Aleatoric and Epistemic Uncertainty with a Single Hybrid Model. *TMLR*. <https://doi.org/10.48550/arXiv.2510.05054>.
- **Van Katwyk et al.** (2025). ISEFlow v1.0: A Flow-Based Neural Network Emulator for Improved Sea Level Projections

and Uncertainty Quantification. *Geoscientific Model Development*. <https://doi.org/10.5194/egusphere-2025-870>.

- **Van Katwyk et al. (2023)**. A Variational LSTM Emulator of Sea Level Contribution from the Antarctic Ice Sheet. *JAMES*. <https://doi.org/10.1029/2023MS003899>.

Additional Publications

- H. Wang, T. Fu, Y. Du, W. Gao, K. Huang, Z. Liu, P. Chandak, S. Liu, **P. Van Katwyk**, A. Deac, A. Anandkumar, K. Bergen, C. Gomes, S. Ho, P. Kohli, J. Lasenby, J. Leskovec, T. Liu, A. Manrai, D. Marks, B. Ramsundar, L. Song, J. Sun, J. Tang, P. Veličković, M. Welling, L. Zhang, C. Coley, Y. Bengio, M. Zitnik (2023). Scientific discovery in the age of artificial intelligence. *Nature*. <https://doi.org/10.1038/s41586-023-06221-2>.
- Seroussi, H., et al. (including Peter Van Katwyk) (2023). Insights into the vulnerability of Antarctic glaciers from the ISMIP6 ice sheet model ensemble and associated uncertainty. *The Cryosphere*. <https://doi.org/10.5194/tc-17-5197-2023>.
- B. C Barton, S. Nelson, J. McBride, B. Bickmore, I. Spring, L. Wells, **P. Van Katwyk**, E. Wolfe (2023). The development of laterite weathering profiles as a function of rainfall and time: A geophysical approach. *Earth Surface Processes and Landforms*. <https://doi.org/10.1002/esp.5688>.

Presentations

1. Using AI to Emulate Antarctic Ice Sheet Contribution to Sea Level Rise. Brown University EEPS Guest Lecture, Feb 2026. Invited Talk.
2. A Broader View of Sea Level Rise: Using ISEFlow to Emulate Antarctic Ice Sheet Response. Brigham Young University Geology Seminar, Jan 2026. Invited Talk.
3. Rewiring Climate Modeling: Integrating Machine Learning Emulators into Earth System Science. AGU 2025. Talk.
4. A Broader View of Sea Level Rise: Using ISEFlow to Emulate AIS Response Across CMIP6. AGU 2025. Talk.
5. Rewiring Climate Modeling with Emulators: A Case Study in Ice Sheet Emulation, Brown University Lunch Bunch Seminar Series, Oct 2025. Talk.
6. Machine Learning emulators of sea level contribution, NASA Sea Level Change Team Seminar, Aug 2025. Invited Talk.
7. Emulation of sea level rise from the Antarctic and Greenland Ice Sheets using ISEFlow, World Climate Research Programme Emulator Task Team, Feb 2025. Invited Talk.
8. Machine Learning emulators of sea level contribution from the Antarctic and Greenland Ice Sheets. Machine Learning for Physical Oceanography Seminar, Jan 2025. Invited Talk.
9. ISEFlow: Emulating Sea Level Rise using a Hybrid Flow-Based Neural Network Architecture. AGU 2024. Poster.
10. AI-ATC: Reinforcement learning for optimal ARTCC sector traffic management. NASA Intern Conference 2024. Talk.
11. Improving Sea Level Projections with AI. Brown University, Research Matters 2024. Talk.
12. Variational LSTM emulators of sea level contribution from the Antarctic and Greenland ice sheets. AGU 2023. Poster.
13. Emulation of ISMIP6 Antarctic sea level contribution and ensemble distributions using time series neural networks. Liege Colloquium on ML and Data Analysis in Oceanography, May 2023. Talk.
14. Normalizing Flows: Uncertainty Quantification in Earth Machine Learning. SAGE GAGE 2021. Poster.
15. Redox and Detrital Geochemical Variation as an Indicator of Organofacies Variability – Heterogeneity, Provenance, and Process in the Mowry Shale, Wind River Basin, Wyoming. AAPG ACE 2020. Poster.
16. Gradational Weathering of Molokai, Hawaii: Geophysical Study of Hawaiian Weathering Profiles. GSA 2020. Poster.

Fellowships and Awards

- WHOI Postdoctoral Fellowship Award, not accepted. 2026.
- National Science Foundation Graduate Research Fellowship (NSF GRFP). 2022-2026.
- US Department of Defense SMART fellowship. Awarded, not accepted. 2021.
- Brigham Young University College of Physical and Mathematical Sciences Dean's List. 2018-2021.
- James & Sally Ellison Geology Scholarship. 2020-2021.
- Alan Hansen Geological Sciences Scholarship. 2019-2020.

Service and Outreach

- **Reviewer:** Nature Climate Change, Journal of Geophysical Research: Machine Learning and Computation
- **Earth Science Teacher & Translator:** Translated Earth Science curriculum to Spanish-speaking students with Brown University DEEPS CORES.